

Unit 1 Test Study Guide

Exponents and Radicals | New Format Sample

Name: _____

Period: _____ Due: _____

How to use this guide: This guide mirrors the Unit 1 test structure while giving each skill a Version A and Version B. You will see 14 multiple-choice practice items and 20 free-response practice items. Try each problem before using the self-check bank at the end.

Multiple-Choice Practice

1A. Power Rule | Version A

Simplify $(-3xy^2)^2$.

- A. $-9x^2y^4$
- B. $9xy^4$
- C. $9x^2y^4$
- D. $-6x^2y^4$

Work / thinking

1B. Power Rule | Version B

Simplify $(2x^2y^3)^3$.

- A. $6x^6y^9$
- B. $8x^6y^9$
- C. $8x^5y^6$
- D. $2x^6y^9$

Work / thinking

2A. Product Rule | Version A

Simplify $-3x^2y^4 \cdot 5x^5y$.

- A. $-15x^7y^5$
- B. $15x^{10}y^4$
- C. $-8x^7y^5$
- D. $-15x^3y^4$

Work / thinking

2B. Product Rule | Version B

Simplify $4a^3b^2 \cdot (-2a^5b^4)$.

- A. $-8a^8b^6$
- B. $8a^{15}b^8$
- C. $-6a^8b^6$
- D. $-8a^2b^2$

Work / thinking

3A. Quotient Rule | Version A

Simplify the quotient shown below. $\frac{12a^5b^7}{3a^2b^3}$ A. $4a^3b^4$ B. $4a^7b^{10}$ C. $9a^3b^4$ D. $4a^2b^3$	Work / thinking
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3B. Quotient Rule | Version B

Simplify the quotient shown below. $\frac{20m^6n^5}{5m^2n^3}$ A. $4m^4n^2$ B. $4m^8n^8$ C. $15m^4n^2$ D. $4m^3n^2$	Work / thinking
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4A. Simplifying Radicals with Variables | Version A

Simplify $\sqrt{72x^4y^3}$. Assume variables are positive. A. $6x^2y\sqrt{2y}$ B. $6x^4y\sqrt{2}$ C. $12x^2y\sqrt{y}$ D. $36x^2y\sqrt{2y}$	Work / thinking
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4B. Simplifying Radicals with Variables | Version B

Simplify $\sqrt{98a^6b^5}$. Assume variables are positive. A. $7a^3b^2\sqrt{2b}$ B. $14a^3b^2\sqrt{b}$ C. $7a^6b^2\sqrt{2b}$ D. $49a^3b^2\sqrt{2b}$	Work / thinking
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5A. Simplifying Radical Quotients | Version A

Simplify the quotient shown below. $\frac{\sqrt{18}}{\sqrt{50}}$ A. $\frac{3}{5}$ B. $\frac{9}{25}$ C. $\sqrt{32}$ D. $\frac{5}{3}$	Work / thinking
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5B. Simplifying Radical Quotients | Version B

Simplify the quotient shown below. $\frac{\sqrt{32}}{\sqrt{8}}$ A. 4 B. 2 C. $\sqrt{24}$ D. $\frac{1}{2}$	Work / thinking
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6A. Adding and Subtracting Radicals | Version A

Simplify $\sqrt{75} + \sqrt{48}$. A. $\sqrt{123}$ B. $9\sqrt{3}$ C. $5\sqrt{3} + 4\sqrt{12}$ D. $13\sqrt{3}$	Work / thinking
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6B. Adding and Subtracting Radicals | Version B

Simplify $\sqrt{45} + \sqrt{20}$. A. $\sqrt{65}$ B. $5\sqrt{5}$ C. $9\sqrt{5}$ D. $13\sqrt{5}$	Work / thinking
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7A. Multiplying Radicals | Version A

Simplify $\sqrt{6} \cdot \sqrt{15}$. A. $\sqrt{21}$ B. $3\sqrt{10}$ C. 90 D. $15\sqrt{6}$	Work / thinking
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7B. Multiplying Radicals | Version B

Simplify $\sqrt{12} \cdot \sqrt{18}$. A. $\sqrt{30}$ B. $6\sqrt{6}$ C. 30 D. $12\sqrt{18}$	Work / thinking
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Free-Response Practice

Show your work clearly. The answer bank is there to help you check, not to replace your process.

8a-A. Evaluating Exponential Expressions | Version A

Evaluate -3^4. Answer: _____	Work / thinking _____ _____ _____ _____
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8a-B. Evaluating Exponential Expressions | Version B

Evaluate -4^3. Answer: _____	Work / thinking _____ _____ _____ _____
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8b-A. Evaluating Exponential Expressions | Version A

Evaluate $(-2)^5$. Answer: _____	Work / thinking _____ _____ _____ _____
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8b-B. Evaluating Exponential Expressions | Version B

Evaluate $(-5)^3$. Answer: _____	Work / thinking _____ _____ _____ _____
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9a-A. Simplifying Radicals | Version A

Simplify $\sqrt{80}$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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9a-B. Simplifying Radicals | Version B

Simplify $\sqrt{108}$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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9b-A. Simplifying Radicals with Variables | Version A

Simplify $\sqrt{(72a^4b^5c^2)}$. Answer: _____	Work / thinking _____ _____ _____ _____ _____ _____
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9b-B. Simplifying Radicals with Variables | Version B

Simplify $\sqrt{(50x^6y^3z^2)}$. Answer: _____	Work / thinking _____ _____ _____ _____ _____ _____
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10a-A. Positive Exponents Only | Version A

Simplify the quotient shown below.

$$\frac{(2x^4y^3)^2}{4x^2y^5}$$

Answer: _____

Work / thinking

10a-B. Positive Exponents Only | Version B

Simplify the quotient shown below.

$$\frac{(3a^3b^2)^2}{9a^2b}$$

Answer: _____

Work / thinking

10b-A. Positive Exponents Only | Version A

Simplify the quotient shown below.

$$\frac{(6m^5n^2)^2}{(3m^2n^4)^2}$$

Answer: _____

Work / thinking

10b-B. Positive Exponents Only | Version B

Simplify the quotient shown below. $\frac{(4p^3q^2)^2}{(2pq^4)^2}$ Answer: _____	Work / thinking
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11a-A. Adding and Subtracting Radicals | Version A

Simplify $8\sqrt{20} - 6\sqrt{5}$. Answer: _____	Work / thinking
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11a-B. Adding and Subtracting Radicals | Version B

Simplify $5\sqrt{45} - 2\sqrt{20}$. Answer: _____	Work / thinking
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11c-A. Multiplying Radicals | Version A

Simplify $2\sqrt{10} \cdot \sqrt{6}$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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11c-B. Multiplying Radicals | Version B

Simplify $3\sqrt{8} \cdot \sqrt{12}$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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11d-A. Dividing Radicals | Version A

Simplify the quotient shown below. $\frac{\sqrt{180}}{\sqrt{5}}$ Answer: _____	Work / thinking _____ _____ _____ _____ _____
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11d-B. Dividing Radicals | Version B

Simplify the quotient shown below. $\frac{\sqrt{288}}{\sqrt{2}}$ Answer: _____	Work / thinking
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11e-A. Rationalizing the Denominator | Version A

Simplify and rationalize the denominator. $\frac{\sqrt{3}}{\sqrt{7}}$ Answer: _____	Work / thinking
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11e-B. Rationalizing the Denominator | Version B

Simplify and rationalize the denominator. $\frac{\sqrt{5}}{\sqrt{11}}$ Answer: _____	Work / thinking
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Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the rule, the sign, and the exponent or radical simplification.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

$3\sqrt{10}$	$4m^4n^2$
$9x^2y^4$	$5\sqrt{5}$
$-8a^8b^6$	$6\sqrt{6}$
2	$7a^3b^2\sqrt{(2b)}$
$-15x^7y^5$	$3/5$
$8x^6y^9$	$9\sqrt{3}$
$6x^2y\sqrt{(2y)}$	$4a^3b^4$

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

$11\sqrt{5}$	-81
a^4b^3	$6\sqrt{3}$
$\sqrt{55/11}$	$4m^6/n^4$
$12\sqrt{6}$	-125
$5x^3yz\sqrt{(2y)}$	$\sqrt{21/7}$
$4\sqrt{15}$	x^6y
6	-64
$4\sqrt{5}$	$4p^4/q^4$
$10\sqrt{5}$	12
-32	$6a^2b^2c\sqrt{(2b)}$

Final Check Before the Test

- ☐ I can explain when to add, subtract, and multiply exponents.
- ☐ I can tell the difference between -3^2 and $(-3)^2$.
- ☐ I can simplify radicals before adding or subtracting.
- ☐ I can simplify radical quotients and rationalize a denominator.
- ☐ I can show enough work for someone else to follow my thinking.